Mixture Densities & EM

A mixture density model:

$$p_\theta(v) = \sum_{j=1}^{M} \alpha_j \, p_{\theta_j}(v)$$

where p_{θ_j} is a density fn.

$0 \le \alpha_j \le 1$ & $\sum_j \alpha_j = 1$

Gaussian Mixture density

$$p_{\theta_j}(\;) = N(\;;\mu_j, \Sigma_j)$$

Maximum likelihood Est. for Mixture

Recall,

$$\theta = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^{N} \log p_\theta(v_i)$$

In the case of mixture density

$$\theta = \{\alpha_j, \mu_j, \Sigma_j\} \quad j = 1 \ldots M.$$

$$\sum_{i=1}^{N} \log p_\theta(v_i) = \sum_{i=1}^{N} \log \sum_{j=1}^{M} \alpha_j \, N(v_i; \mu_j, \Sigma_j)$$

Latent Variable Models.

Given $D = \{x_1, x_2 \ldots x_N\} \sim \text{iid } \mathbb{P}_x$

$$x_i \in \mathbb{R}^d$$

Introduce another RV into the system that is un-observed. (hidden / latent RV)

Mathematically, $\forall x_i \in D$, define $z_i \in \mathbb{R}^K$ $K \ll d$ with a distribution of its own.
X & Z are assumed to be correlated.

Z can be continuous / discrete.

With this a latent var. model is defined as

$$p_\theta(x) = \sum_z p_\theta(x, z) \quad \text{if } Z \text{ is discrete}$$

$$= \int_z p_\theta(x, z)\, dz \quad z \text{ is conti.}$$

ML Estimation for Latent variable Models.

For a lat. variable model both the Model parameters & the density over the lat. var needs to be estimated.

We have the likelihood function $l(\theta)$

$$l(\theta) = \log p_\theta(x)$$

$$= \log \sum_z p_\theta(x, z)$$

Suppose $q(z)$ is some density over z.

$$l(\theta) = \log \sum_z \left[p_\theta(x,z) \cdot \frac{q(z)}{q(z)} \right]$$

$$= \log \sum_z q(z) \cdot \underbrace{\left(\frac{p_\theta(x,z)}{q(z)} \right)}_{\text{fn. of } z}$$

$$= \log \mathbb{E}_{q(z)} \left[\frac{p_\theta(x,z)}{q(z)} \right]$$

Using Jensen's inequality & noting that log is a concave function,

$$\underbrace{\log \mathbb{E}_{q(z)} \left[\frac{p_\theta(x,z)}{q(z)} \right]}_{l(\theta)} \geq \underbrace{\mathbb{E}_{q(z)} \log \left[\frac{p_\theta(x,z)}{q(z)} \right]}_{F_\theta(q)}$$

$$l(\theta) \geq F_\theta(q)$$

$F_\theta(q)$ is now computable $\because$ it has "sum of log" type terms.

$F_\theta(q)$ is a lower bound on $\ell(\theta)$

log likelihood $\leftarrow$ Evidence.

Evidence Lower bound (ELBO).

New optimization problem:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \; \ell(\theta) \quad : \text{original}$$

$$\hat{\theta}^*, q = \underset{\theta, q}{\operatorname{argmax}} \; F_\theta(q) \quad : \text{modified optimization}$$

Question : What choice of $q(z)$ would make the ELBO tight?

Consider

$$\ell(\theta) - F_\theta(q) = \log p_\theta(x) - \sum_z q(z) \log \frac{p_\theta(x,z)}{q(z)}$$

$$= \log p_\theta(x) - \sum_z q(z) \log \frac{p_\theta(x) \cdot p_\theta(z|x)}{q(z)}$$

$$= \cancel{\log p_\theta(x)} - \cancel{\log p_\theta(x)} - \sum q(z) \log \frac{p_\theta(z|x)}{q(z)}$$

$$= \sum_z q(z) \log \frac{q(z)}{p_\theta(z|x)}$$

$$= D_{KL}\left(q(z) \,||\, p_\theta(z|x)\right)$$

$$= \ell(\theta) - F_\theta(q)$$

$\max F_\theta(q) = \ell(\theta)$, by construction.

$\therefore \ \ell(\theta) - F_\theta(q) = 0$ iff $D_{KL}\left(q(z) \,||\, p_\theta(z|x)\right) = 0$

$$\boxed{\Rightarrow q^*(z) = p_\theta(z|x).}$$

An iterative Algorithm for optimizing $\ell(\theta)$.

Initialize θ'

For $t = 1$ to convergence:

do:

Compute $q^t(z) = p_{\theta^t}(z|x)$

Estimate $F_\theta(q^t) = \mathbb{E}_{q^t(z)} \log \frac{p_\theta(x,z)}{q^t(z)}$

$\hookrightarrow$ Expectation step

Estimate

$$\theta^{t+1} \leftarrow \underset{\theta}{\operatorname{argmax}} \left[F_\theta(q^t) \right]$$

$\hookrightarrow$ maximization step.

To show EM guarantees

$$\ell(\theta^{t+1}) \geq \ell(\theta^t).$$